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D214: Capstone

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Capstone Executive Summary

Problem and Hypothesis

This research project sought to investigate a potential relationship between NFL positional spending and team performance. NFL teams are limited in the amount they can pay their players by a team salary cap, meaning they must budget player salaries to stay below the cap. Star quarterbacks generally sign massive contracts when they complete their rookie contract, some of which can be as large as 20% of the team's salary cap.¹ This project was motivated by the theory that quarterback contracts are disproportionately large, and come at the expense of a strong offensive line, which could in turn lead to a higher injury risk factor for the quarterback. The goal of the project was to first identify if there was a measurable relationship between quarterback and offensive line salaries, and then identify if there was a measurable relationship between offensive line salaries and quarterback injuries. After this, the premise was expanded to more broadly attempt to predict end of season win-loss records for NFL teams based on how the team budgeted its salary cap around each position.

The research question of the project was as follows: ***can an analysis of NFL player salaries be used to predict team performance (win-loss records)?***

The hypothesis of the project was as follows: ***There is a statistically significant correlation between NFL position salaries and win-loss records.***

Data Analysis

The following variables were collected using web scraping, and cleaned with python code. These variables were collected for every NFL team for each year from 2014 to 2023 (excluding the 2021 Dolphins, and Chargers, which could not be obtained due to errors from the data source).

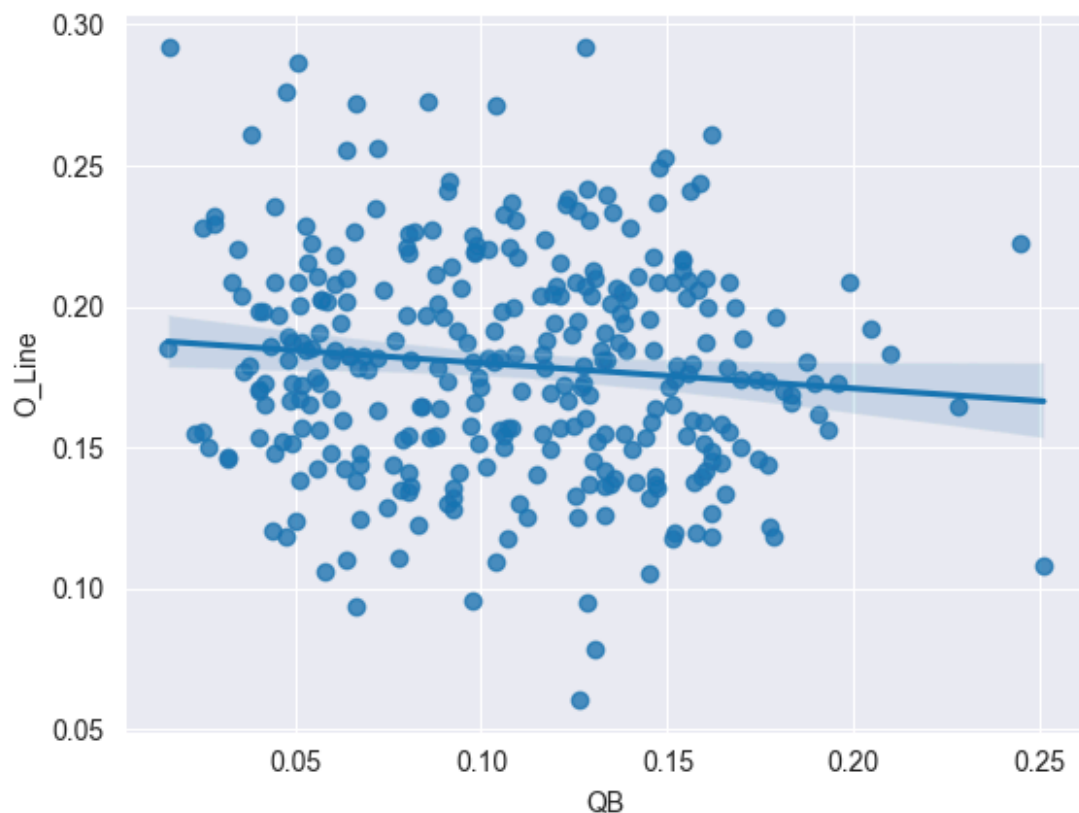
- Team Name
- Year
- Win-Loss Record
- Strength of Schedule

- Number of weeks that the usual starting quarterback missed due to injury
- Total salaries grouped by each position on the roster, represented as a percentage of the team's total spending for that year

After cleaning the data was split into 80% training data and 20% testing data. First, a linear model was trained on the total offensive line spending as a function of quarterback spending. Next, a logistic model was trained on whether the team lost a quarterback to injury as a function of total offensive line spending. Finally, a linear model was trained on the NFL team's win-loss record as a function of its strength of schedule and the team's overall salary budget (broken down by position). These models were trained using the ols and logit functions from the python library statsmodels. Each model was then evaluated by predicting values with the testing data set. The linear models were evaluated using their R-squared values and residual standard errors (RSE), and the logistic model was evaluated using its precision, recall, pseudo-R-squared value, and p-value. As an exercise, the last model (win-loss record as a function of spending) was used to predict the end of season records for the current 2024 NFL teams.

Overall Findings

None of the models were able to return significant correlations in order to reject the null hypothesis.



The first linear model had an R-value of -0.011 and an RSE that was 22.% of the mean when used on the testing data. This strongly indicates the lack of a linear correlation between the two variables. A scatterplot is shown below, which illustrates the lack of a clear linear trend:

The logistic model had precision and recall values that were both zero. The pseudo-R-squared

Confusion Matrix		
	predicted false	predicted true
actual false	56	0
actual true	8	0

value was 0.0003 (too low to indicate a logistic correlation), and the p-value was 0.815 (too high to reject the null hypothesis). The confusion matrix shown below illustrates the failure of the model to predict any positive values.

The second linear model had an R-squared value of -0.092 and an RSE that was 35.4% of the mean when used on the testing data. Again, the extremely low R-squared value, and high RSE value indicate an inability to identify linear correlations between the independent and dependent variables used to fit the model.

While the models failed to reject the null hypothesis, end of season records were still predicted for each team (noting that the results of the prediction should not be trusted for unsuccessful models). The model selected the Chicago Bears to have the best end of season record of 0.678, and the Green Bay Packers to have the worst end of season record at 0.280.

Limitations

The overall linear model (used to predict end of season records) was significantly limited by the inability to consider interaction terms between variables. The model was trained on 26 variables (strength of schedule plus 25 positional salary variables), meaning there are 351 interactions to consider ($\# \text{ interactions} = n(n+1)/2$). Since there are only 32 NFL teams, there are only 32 data points per year, meaning there is simply not enough data to train a model with this many interactions. The linear model is also limited by the inability to detect non-linear trends.

The data collection process was also limited by the depth of data selected. For example, by not considering the play time and injury status, this introduced a confounding variable by

adding additional weight to a player's salary if they did not play. Higher paid offensive linemen that missed significant time due to injury were not able to as significantly affect end of season results, meaning their salary was weighted too significantly in the analysis.

Proposed Actions

While the analysis did not result in any of the models showing statistically significant correlations, it is still possible that a model could be created that is successful in rejecting the null hypothesis. I recommend expanding on the model through more rigorous data collection. This would include considering potential relationships between salaries and in game statistics (for example: do higher paid offensive lines allow less sacks?) as well as considering injuries and playtimes at all positions, as discussed previously.

Expected Benefits

The expected benefit of this study, were it to successfully reject the null hypothesis, is that it would allow NFL teams the ability to more strategically plan their team budgets by position. If a model were successfully created, it would allow teams to optimize the amount of money they spend at each position, and potentially increase their chances for better end of season results by identifying undervalued players to sign at certain positions in order to afford high valued players at other positions (this is a strategy that has long been used in baseball called "moneyball"²).

Sources

¹ A historical table of quarterback contracts can be found at <https://overthecap.com/contract-history/quarterback>.

² An explanation of moneyball in baseball can be found at <https://eutawstreetreport.com/the-impact-of-moneyball-in-mlb/>.